Seok-Jin Kang

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Bibliography

Seok-Jin Kang, Ph.D. is an **immunologist** and **computational biologist** whose academic and research trajectory bridges traditional immunology and modern **data-driven biology**. During his M.S. and Ph.D. training in the **Immune Modulation Laboratory** *(PI: Prof. Taehoon Chun)* at **Korea University**, he investigated **immune regulation**, **macrophage polarization**, and the **molecular mechanisms** of **host–pathogen interactions** using both ***in vitro*** and ***in vivo*** models.

He strategically pivoted from wet-lab immunology to computational biology to overcome the limitations of experimental methods in capturing a systems-level understanding of complex immune networks. Recent work focuses on the **modeling and identification of intrinsically disordered regions (IDRs)** using **quantum computing**, as well as applying **computer-aided drug design (CADD)** to uncover the **biophysical mechanisms** underlying **protein function and regulation**.

In the short term, his research aims to advance **precision medicine** through the development of **multi-omics-based computational models** capable of predicting **immune responses**, guiding **therapeutic design**, and enabling **personalized intervention strategies**. Over the long term, he aspires to establish himself as an **independent faculty member** and contribute to extending **human healthspan and wellbeing** through integrative, **AI-driven biomedical research** that unites **immunology**, **genomics**, and **computational modeling**.

Research Interests

Precision Medicine, CAR-T Cell Therapy, Multi-omics Integration, Computer-aided Drug Design (CADD), Quantum Computing

Careers

Korea University (2026 QS ranking: 61st) **Mar 2026 – Present**
Research Professor, Immune Modulation Laboratory (Advisor: Prof. Taehoon Chun)
Korea University **Mar 2025 – Feb 2026**
Postdoctoral Fellow, Immune Modulation Laboratory (Advisor: Prof. Taehoon Chun)

Education

Korea University (2026 QS ranking: 61st) **Mar 2018 – Feb 2025**
M.S. & Ph.D. in Biotechnology
Immune Modulation Laboratory (Advisor: Prof. Taehoon Chun)

Korea University **Mar 2014 – Feb 2018**
B.S. in Biotechnology

Publications

*† indicates first author; * indicates corresponding author.*

Featured Publications

5. ***SJ Kang*†**, H Shin*, Biophysical mechanisms of spider-silk constituting element–induced stick-slip behavior and hydrogen bond regeneration for high toughness in silk fibers, *International Journal of Biological Macromolecules*, 147027, 2025 (Impact factor : 8.5, Q1)
4. ***SJ Kang*†**, H Shin*, Amino acid sequence-based IDR classification using ensemble machine learning and quantum neural networks, *Computational Biology and Chemistry*, 108480, 2025 (Impact factor : 3.1, Q1)
3. ***SJ Kang*†**, J Yang†, NY Lee, CH Lee, IB Park, SW Park, HJ Lee, HW Park, HS Yun, T Chun*, Monitoring cellular immune responses after consumption of selected probiotics in immunocompromised mice, *Food Science of Animal*

Resources 42 (5), 903, 2022 (Impact factor : 3.7, Q2)

2. ***SJ Kang***†, IB Park, T Chun*, Open reading frame 5 protein of porcine circovirus type 2 induces RNF128 (GRAIL) which inhibits mRNA transcription of interferon-β in porcine epithelial cells, *Research in Veterinary Science* 140, 79–82, 2021 (Impact factor : 1.8, Q2)
1. ***SJ Kang***†, T Chun*, Structural heterogeneity of the mammalian polycomb repressor complex in immune regulation, *Experimental & Molecular Medicine* 52 (7), 1004–1015, 2020 (Impact factor : 12.9, Q1)

Under Review

- ***SJ Kang***†, H Shin*, Quantum-Enhanced Weighted Gene Co-expression Network Analysis Reveals Regulatory Networks Underlying Sexual Size Dimorphism in *Macrobrachium nipponense*, *Integrative Zoology*, under review (Round 1), 2026 (Impact factor : 3.7, Q1)
- ***SJ Kang***†, H Shin*, Quantum-Enhanced Transfer Learning for IDR Binding Partner Prediction, *IEEE Transactions on Computational Biology and Bioinformatics*, under review, 2025 (Impact factor : 3.4, Q1)

Published Articles

- SW Park†, CY Choi†, IB Park†, ***SJ Kang***, HJ Lee, JI Kim, J Bae, D Geum, YP Cheon, T Chun*, LDB1 represses fetal hemoglobin expression by enhancing BCL11A transcription, *Redox Biology*, 118, 108480, 2026 (Impact factor : 11.9, Q1)
- JH Yoon†, GB Yeon†, H Lee, H An, J Oh, ***SJ Kang***, IB Park, SA Lim, SS Hwang, DS Kim, JH Kim, T Chun, YX Fu, J Bae*, Tumor-targeted delivery of CXCL10 by mesenchymal stromal cells potentiates adoptive T cell therapy to treat solid tumors, *Biomedicine & Pharmacotherapy* 192, 118579, 2025 (Impact factor : 7.5, Q1)
- CH Lee†, HJ Lee†, SW Park, J Shin, ***SJ Kang***, IB Park, HK Kim, T Chun*, Mutational analysis of pig tissue factor pathway inhibitor α to increase anti-coagulation activity in pig-to-human xenotransplantation, *Biotechnology Letters* 46 (4), 521–530, 2024 (Impact factor : 2.1, Q3)
- SW Park†, IB Park†, ***SJ Kang***, J Bae, T Chun*, Interaction between host cell proteins and open reading frames of porcine circovirus type 2, *Journal of Animal Science and Technology* 65 (4), 698, 2023 (Impact factor : 3.2, Q1)
- SP Choi†, SW Park, ***SJ Kang***, SK Lim, MS Kwon, HJ Choi, T Chun*, Monitoring mRNA expression patterns in macrophages in response to two different strains of probiotics, *Food Science of Animal Resources* 43 (4), 703, 2023 (Impact factor : 3.7, Q2)
- CY Choi†, CH Lee, J Yang, ***SJ Kang***, IB Park, SW Park, NY Lee, HB Hwang, HS Yun, T Chun*, Efficacies of potential probiotic candidates isolated from traditional fermented Korean foods in stimulating immunoglobulin A secretion, *Food Science of Animal Resources* 43 (2), 346, 2023 (Impact factor : 3.7, Q2)
- SP Choi†, YC Choi, J Yang, CY Choi, CH Lee, ***SJ Kang***, IB Park, T Chun*, Monitoring mRNA transcription of genes involved in early pregnancy from endometrium and peripheral blood mononuclear cells of pregnant pigs with different parity, *Reproduction in Domestic Animals* 53 (6), 1594–1599, 2018 (Impact factor : 1.7, Q3)
- CY Choi†, YC Choi, IB Park, CH Lee, ***SJ Kang***, T Chun*, The ORF5 protein of porcine circovirus type 2 enhances viral replication by dampening type I interferon expression in porcine epithelial cells, *Veterinary Microbiology* 226, 50–58, 2018 (Impact factor : 2.7, Q1)

Patents

Registration

- T Chun, J Yang, CH Lee, ***SJ Kang***, IB Park, Chimeric antigen receptor specifically binding to CD38 and use thereof, *Korean Intellectual Property Office*, 10-2021-0025895, October 2021
- T Chun, J Yang, CH Lee, ***SJ Kang***, IB Park, Chimeric antigen receptor specifically binding to VLA-4 and use thereof, *Korean Intellectual Property Office*, 10-2021-0025905, March 2021

Pending

- ***SJ Kang***, H Shin, Method and system for classifying intrinsically disordered proteins based on amino acid sequences using quantum neural networks, *Korean Intellectual Property Office*, 10-2025-0078871, June 2025
- T Chun, IB Park, ***SJ Kang***, HJ Lee, Regulation of BCL11A expression by LDB1 transcriptional regulator and uses thereof, *Korean Intellectual Property Office*, 10-2025-0010845, January 2025

- T Chun, IB Park, ***SJ Kang***, HJ Lee, Regulation of PLZF expression by LDB1 transcriptional regulator and uses thereof, *Korean Intellectual Property Office*, 10-2025-0010844, January 2025

Certifications

- **Big Data Analysis Engineer**, *Korea Data Agency*, 2025, BAE-010002862
- **ADsP (Advanced Data Analytics Semi-Professional)**, *Korea Data Agency*, 2024, ADsP-040004911

Research Projects

13. **Q2431261** *02/2025 – 12/2025*
    Title: Safety evaluation of viral vector-based gene therapy with respect to non-specific genome insertion
    Source: Ministry of Food and Drug Safety
    Role: Research Assistant *(PI: Prof. Taehoon Chun)*
    - **Vector engineering** (Lentiviral/AAV) and **in vivo** administration
    - **WGS/NGS library construction** for genomic integration profiling
    - **Statistical analysis** of insertion site distribution and toxicity correlations

12. **Q2515581** *05/2025 – 12/2025*
    Title: Optimization of HLA gene editing for the development of allogeneic stem cell therapy
    Source: Korea National Institute of Health
    Role: Research Assistant *(PI: Prof. Dongho Geum)*
    - **Hypoimmunogenic iPSC generation** via CRISPR-Cas9 HLA editing
    - **Directed differentiation** into immune-evasive cardiomyocytes and T cells
    - ***In vivo* engraftment validation** using humanized murine models

11. **R2222761** *09/2022 – 02/2025*
    Title: Regulation of immune cell differentiation by chromatin 3D structural changes
    Source: National Research Foundation of Korea
    Role: Research Assistant *(PI: Prof. Taehoon Chun)*
    - **Grant Writing:** Multi-omics pipeline structuring and statistical planning
    - **Chromatin 3D structural analysis** mediated by LDB1 looper
    - **Multi-omics integration** (RNA/ATAC/ChIP-seq) for enhancer–promoter mapping
    - ***In vivo* validation** using LDB1-floxed mouse models

10. **R2212861** *04/2022 – 12/2023*
    Title: Development of universal SARS-CoV-2 and sarbecovirus vaccine candidate using NDV vector platform
    Source: Korea Health Industry Development Institute
    Role: Research Assistant *(PI: Prof. Kisoon Kim)*
    - **Grant Writing:** Proposal submission and **AlphaFold application strategy**
    - **Structural epitope modeling** using AlphaFold for cross-protection
    - **Immunogenicity evaluation** in viral challenge animal models
    - **Immune profiling** via ELISA and flow cytometry

9. **R2131151** *02/2022 – 12/2025*
    Title: Development of cellular immune evaluation technology for viral vector-based genetic vaccines
    Source: Ministry of Food and Drug Safety
    Role: Research Assistant *(PI: Prof. Taehoon Chun)s*
    - **Candidate selection** for Adenovirus/Vaccinia-based platforms
    - **Assay development** for humoral and cellular immunity assessment
    - **Safety profiling** including cytotoxicity and organ-specific toxicity

8. **R1923081** *09/2019 – 02/2022*
    Title: Mechanistic study of macrophage polarization regulated by chromatin remodeling factor Phc2
    Source: National Research Foundation of Korea
    Role: Research Assistant *(PI: Prof. Taehoon Chun)*

- **Transcriptomic profiling** of macrophage polarization states
- **Differential expression analysis** (DEG) and pathway enrichment
- **Regulatory network identification** of Phc2-mediated activation

7. **R1727591** *01/2018 – 12/2020*
   Title: Development of DNA markers to enhance resistance against chronic wasting diseases
   Source: Rural Development Administration
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - **SNP genotyping** and sequence analysis for resistance markers
   - **Pipeline development** using Python/Biopython for variant detection

6. **R1715822** *03/2018 – 06/2022*
   Title: Development of novel immune cell therapy targeting multiple myeloma
   Source: National Research Foundation of Korea
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - **CAR construct engineering** and optimization via molecular cloning
   - **Binder selection** for high affinity and specificity
   - **Kinetics analysis** (K_d) using **surface plasmon resonance (SPR)**

5. **Q2029461** *01/2021 – 06/2023*
   Title: Production and efficacy evaluation of novel recombinant immunotherapeutics
   Source: Immunological Designing Lab Co., Ltd.
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - **Recombinant protein production** and purification
   - **Affinity characterization** via SPR and ELISA

4. **Q2015891** *06/2020 – 02/2021*
   Title: Screening of immune-enhancing probiotics using immunodeficient mouse models
   Source: CJ CheilJedang & BLOSSOM PARK
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - ***In vivo* screening** and cytokine profiling (Flow cytometry/qPCR)
   - **Statistical evaluation** of immune enhancement using R

3. **Q1815701** *06/2018 – 11/2018*
   Title: Determination of dosing period and amount of probiotics for alleviating allergic rhinitis
   Source: CJ CheilJedang
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - **Th1/Th2 cytokine profiling** via ELISA and gene expression
   - **Dosing optimization modeling** for immune regulation

2. **PJ013271012018** *03/2018 – 12/2020*
   Title: Development of genetic markers associated to the resistance to PMWS
   Source: Rural Development Administration
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - **Comparative genomics** for resistance marker identification
   - **Variant detection pipeline** using BLAST and Biopython

1. **Q1613841** *07/2016 – 04/2017*
   Title: Selection and *in vivo* efficacy verification of probiotics with enhanced IgA secretion
   Source: CJ CheilJedang
   Role: Research Assistant *(PI: Prof. Taehoon Chun)*
   - **Mucosal IgA quantification** in murine models
   - **Immune gene validation** using qPCR

Book chapters

International Publications

- ***SJ Kang***, **2026 Science Trends, Chapter: Multi-omics**, *Kindle*, ISBN 9798266588004, September 2025
- ***SJ Kang***, **2025 Science Trends, Chapter: Artificial Intelligence**, *Kindle*, ISBN 979-8307079270, January 2025

Domestic Publications

- ***SJ Kang***, **2026 생명과학트렌드, 챕터: 멀티오믹스(2026 Life Science Trends, Chapter: Multi-omics)**, *Pubple*, ISBN 9788924169034, August 2025
- ***SJ Kang***, **2026 과학트렌드, 챕터: 개인 맞춤형 치료(2026 Science Trends: Personalized Medicine)**, *Pubple*, ISBN 9788924169041, August 2025
- ***SJ Kang***, **2025 과학트렌드, 챕터: 인공지능(2025 Science Trends, Chapter: Artificial Intelligence)**, *Pubple*, ISBN 9788924143621, December 2024